Software Requirement Specification Document for the Bank of Jarvis Online Investment Application

Version 1.0

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0. Table of Contents

0. Table of Contents
- Revision History

1. Introduction
- 1.1 Background
- 1.2 Project Scope
- 1.3 Intended Audience & Reading Suggestions

2. Overall Description
- 2.1 Product Perspective
- 2.2 Product Features
- 2.3 User Classes and Characteristics
- 2.4 Operating Environment
- 2.5 Design and Implementation Constraints
- 2.6 User Documentation
- 2.7 Assumptions and Dependencies

3. System Features
- 3.1 Stock Trading
  - 3.1.1 Description/Priority
  - 3.1.2 Stimulus/Response Sequences
  - 3.1.3 Functional Requirements
- 3.2 User Login
  - 3.2.1 Description/Priority
  - 3.2.2 Stimulus/Response Sequences
  - 3.2.3 Functional Requirements
- 3.3 User Portfolio
  - 3.3.1 Description/Priority
  - 3.3.2 Stimulus/Response Sequences
  - 3.3.3 Functional Requirements

4. External Interface Requirements
- 4.1 User Interfaces
- 4.2 Hardware Interfaces
- 4.3 Software Interfaces
- 4.4 Communications Interfaces

5. Other Nonfunctional Requirements
- 5.1 Performance Requirements
- 5.2 Safety Requirements
- 5.3 Security Requirements
- 5.4 Software Quality Attributes

6. Other Requirements
Appendix A: Glossary
Appendix B: Issues List
Appendix C: References

Revision History

Name	Date	Reason for Change	Version

1. Introduction

1.1 Background

The Bank of Jarvis wishes to launch an online investing application to stay competitive with competitors who have also begun to roll out their own. Currently, customers must go through the tedious process of booking an appointment with a stockbroker at a local Bank of Jarvis service branch to buy or sell stocks. This means they need to find time to be available during a busy workweek to place a market order, which is a significant inconvenience for our customer base. For the sake of customer retention, there is a strong push to have an online stock trading service within the fiscal year.

1.2 Project Scope

The Bank of Jarvis' Online Investment application should be ready for release within eight months of development, and capable of servicing over 150,000 Bank of Jarvis' customers. The application is responsible for providing an interface on web and mobile platforms for limited access to the Bank of Jarvis' customer database. The interface will allow customers to view their stock transactions, their stock portfolio, create market orders, and buy and sell stocks over the course of a workday. Any market orders or requests to buy and sell stocks should be resolved instantly and follow ACID Protocols, though the actual buying and selling of stocks will be performed at the end of the day. For the 1.0 release, the application will only support market orders, and other investment options must be done in person at a service branch.

From a technical standpoint, the application will use the Bank of Jarvis' database to access user information and market order transactions and will be only responsible for generating requests to retrieve data from the database and transactions to store market orders to the database as a batch store.

1.3 Intended Audience & Reading Suggestions

This living document was designed for developers, project managers, and documentation writers for the version 1.0 Bank of Jarvis Online Investing Application. The document contains an overview of the project's context, features, and requirements. It's organized so that the reader will be given a brief background of the project, what we intend to build, and the considerations that we have accounted for in order to meet project expectations.

If you are a developer, focus on sections 2 and 3, where most of the feature requirements exist, and skim section 4 for any inquiries about hardware and software expectations.

If you are a project manager, read Section 2, and skim Sections 3, 4, and 5 to get an overall understanding of the project.

If you are a documentation writer, Sections 4 and 5 will provide information about the hardware and user interface experiences. Be sure to review the revision history to see if there are changes to the scope or functional requirements.

2. Overall Description

2.1 Product Perspective

The Bank of Jarvis Online Investment App is a follow-on to the Bank of Jarvis' mobilization towards online convenience for their customers. As their competitors have already implemented mobile investing into their ecosystem, the Bank of Jarvis wishes to remain competitive and provide similar services to their customers. Instead of having to call or go in person to one of its branches, a customer can make requests to buy and sell stocks online and have them processed at the end of the day.

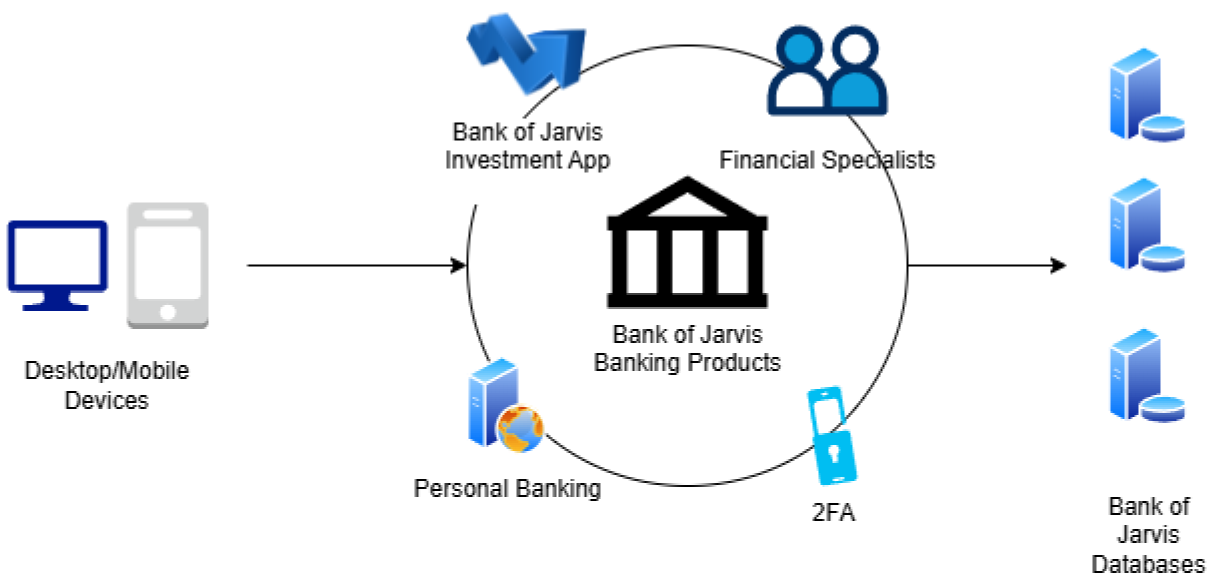

Figure 1: Bank of Jarvis' Product Ecosystem

2.2 Product Features

Customers of the Bank of Jarvis will be able to use this application in order to make day-to-day market orders instead of needing to book a time to do so with a stock broker at a local service branch. They will be able to view their funds, stock portfolio, as well as a live feed of the stock's value updated at most every 5 minutes. Customers will also be able to make market orders using the app by searching a stock up by ticker number or company name. In order to access their account, customers will need to activate their online investment account at a local service branch, create a separate password from their online banking app, and set up 2FA if they have not done so.

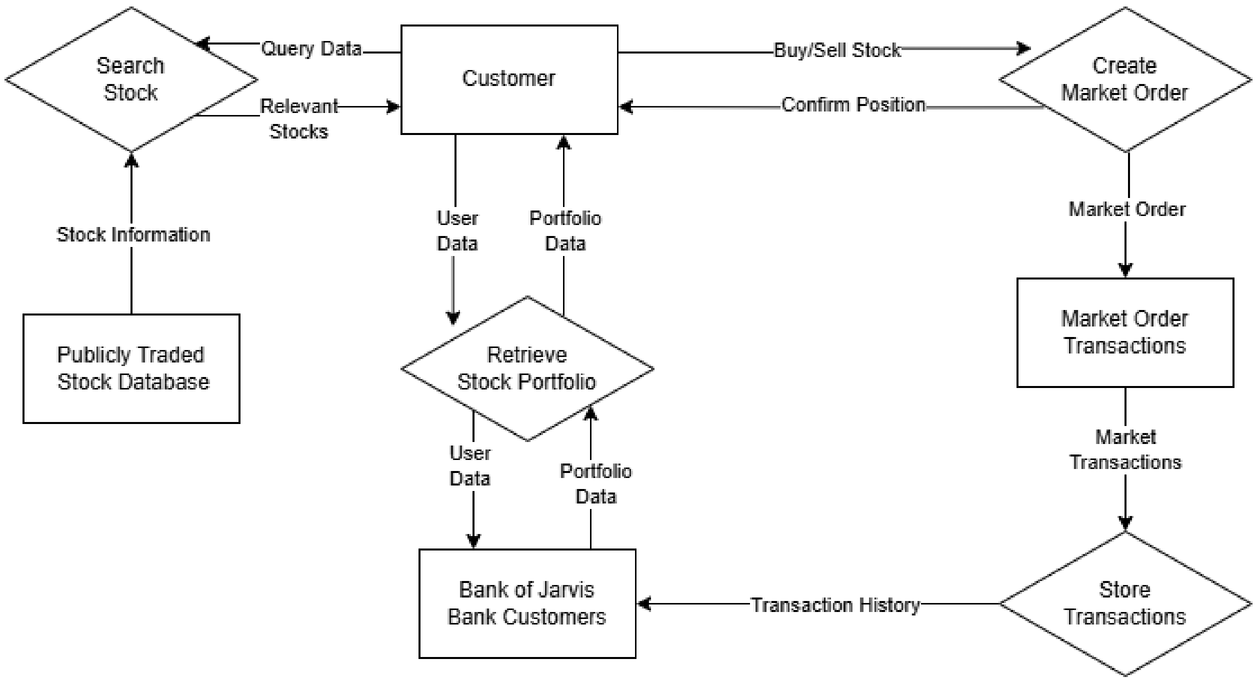

Figure 2: Dataflow of Primary Customer Experience

2.3 User Classes and Characteristics

The application's intended users are Bank of Jarvis customers, and made with the intent to service their small to medium sized business customers. Users are also expected to already have an account with the Bank of Jarvis, and will need to activate their account for online investing at their local service branch in order to access this application.

The Bank of Jarvis expects users to access the app up to four to five times a day, but there will be no hard limit enforced. However, positions will not be updated in real-time and will be processed by the end of the day. Due to this implementation detail, the application does not support day trading.

The application will implement a simple and limited UI to be more accessible to users inexperienced with IoT applications, as well as those who are new to stock trading. Since we are also only focused on market orders, this may dissuade experienced investors looking to create consolidated portfolios.

2.4 Operating Environment

The app will be made publicly available on the web, Android, and iOS. For the web application, devices must be able to run modern web browsers such as Firefox, Google Chrome, Microsoft Edge, or Apple Safari. Due to security risks of older mobile operating systems, we will only support devices that run Android 13 and iOS 16 and later. A majority of the market[3] uses Windows and Mac OS as their desktop operating system, so web support will be focused on those two operating systems.

The hardware requirements for each environment is listed below:

Web[4]:

- Minimum CPU: Intel Core i3 or AMD Ryzen 3
- Minimum OS: Windows 10 version 1607 or later, Mac OS X 10 or later
- Minimum RAM: 4GB

Android[2]:

- Minimum CPU: Snapdragon 8 Gen 2
- Minimum OS: Android 13 or later
- Minimum RAM: 4GB

iOS[1]:

- Minimum CPU: A15 Bionic
- Minimum OS: iOS 16 or later
- Minimum RAM: 4GB

2.5 Design and Implementation Constraints

This application is a limited interface for the Bank of Jarvis' user database, and as such, requires a steady internet connection to function. The hardware must be able to run either web browsers, .ipa, or .apk files to use the interface. 2.4 lists the minimum hardware and OS requirements needed to avoid any security issues.

Since the application must be completed within a short time constraint, using an industry-standard framework will save the most time for development. For solely web applications, React, Vue.js, or Angular are industry-standard frameworks that can be considered. For mobile development, React Native and Xamarin allow cross-platform development between iOS and Android, while Flutter and Ionic allow development for both web and mobile development. Separating the development between web and mobile development can keep the environments separate, but it will depend on how the team handles the development environment.

Note that this application communicates sensitive information with the Bank of Jarvis database; all data sent and received must be encrypted. The Bank of Jarvis will be responsible for maintaining and running the database, and any changes will require contacting the team responsible for creating the database.

Finally, the mobile applications must be Android Google Play and iOS App Store compliant. This is to ensure they are distributable for the Bank of Jarvis' customer base. Customers who use the mobile app are responsible for updating the app when a new major version is released. The web application should be hosted alongside the other Bank of Jarvis' online services on a cloud network to minimize downtime and for the developers to focus on implementing the interfaces.

2.6 User Documentation

As this application is a follow-on to the Bank of Jarvis' online services, and thus, user manuals, online help, and basic tutorials can be accessed on the Bank of Jarvis website. Upon a user's first login into the application, there will be a small walkthrough on how to generally use the app for each supported platform. The terms of service will also be available on the website, and as a link to the web page within the application.

2.7 Assumptions and Dependencies

As the application depends heavily on the Bank of Jarvis database, the developers assume that it will be able to verify and process API requests, store and process transactions, and allow limited access to customer-sensitive data on a need-to-know basis. It also expects an accurate response value that indicates whether a request is properly received. The application will also assume that the database will have a method to provide transactions still yet to be stored in the database, to allow users to verify they have made a transaction, but have yet to be fully stored within the database.

In terms of security, the application also assumes that users will have already set up 2FA for their bank account and that the application will be able to share the same 2FA security service as with the rest of the Bank of Jarvis' online services. These 2FA services will be assumed to be robust and secure enough for use across the different online services that the Bank of Jarvis provides.

3. System Features

3.1 Stock Trading

3.1.1 Description/Priority

This is a high priority core feature that the customers will expect on release. The application cannot be pushed out if this feature is not robust and well-tested.

3.1.2 Stimulus/Response Sequences

1. If a user presses on a “Market Orders” icon, they should be directed to a page with a condensed list of their currently owned stocks and their current market values, as well as a search bar to look up other stocks.
2. If a user presses a search bar icon, it should allow the user to type in an alphanumeric value and see stocks that have a similar ticker number or company name to the inputted value.
3. If a user presses on a search result, they should see the current stock’s price, and buttons to buy or sell the stock if they have sufficient funds or shares of that stock respectively. The buttons should appear disabled if they do not meet the requirements.
4. If a user presses a Buy or Sell button on a stock, they should be prompted to confirm the number of stocks they wish to buy or sell.
    a. If a user enters the number of stocks, they should immediately see the amount of liquid assets or available positions they have and whether or not they can buy a stock or sell a position.
5. If a user has a valid buy or sell request, and they confirm the request, they will be shown a second window chance to confirm or cancel this request.
6. If a user presses “Confirm”, they will be shown whether the transaction is successful or not. If it was not successful, they will be notified that the transaction will not be processed.
7. If a user presses “Cancel”, they will be redirected back 4.a.

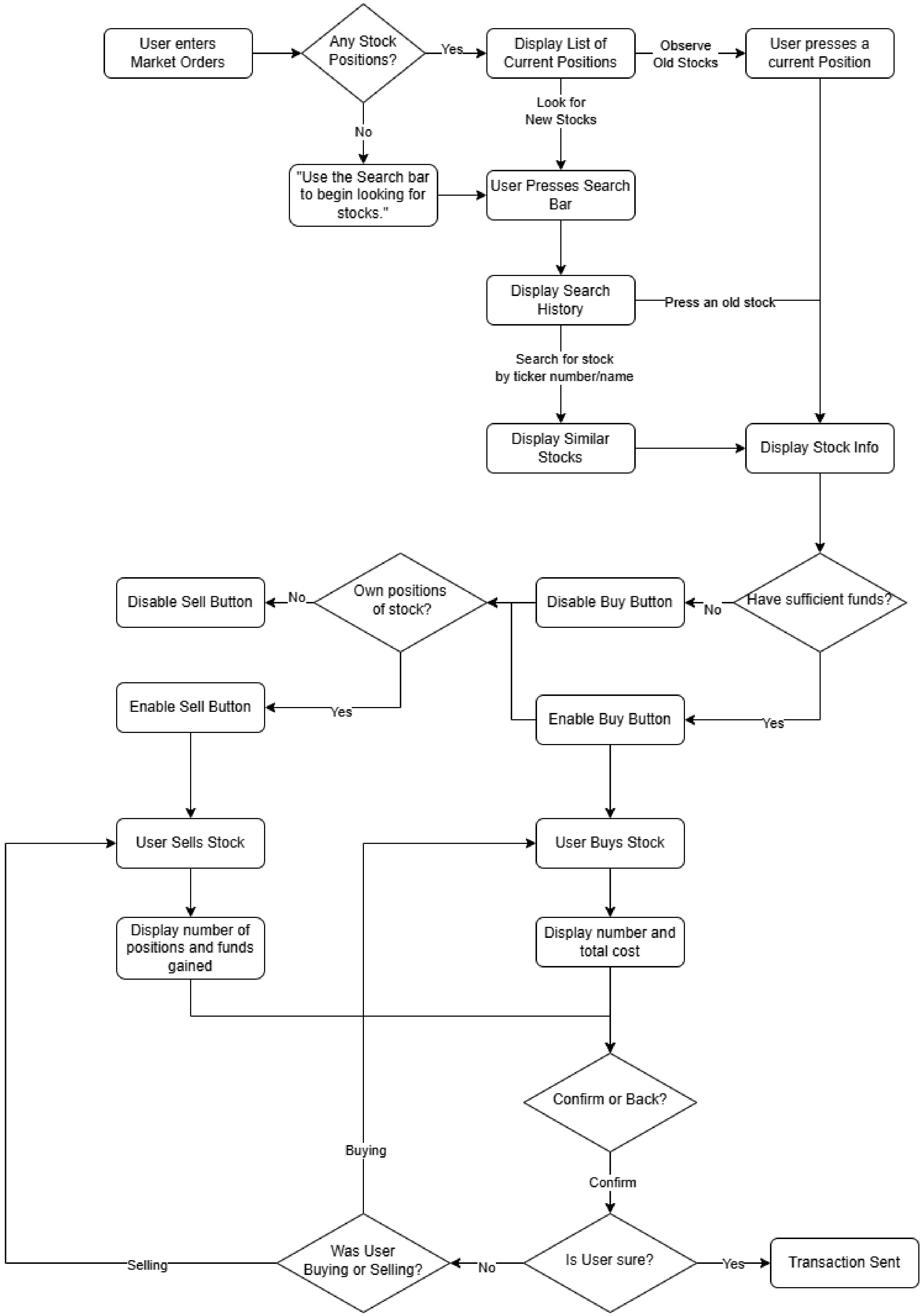
User enters Market Orders
Any Stock Positions?
Yes
No
Display List of Current Positions
Observe Old Stocks
User presses a current Position
Look for New Stocks
"Use the Search bar to begin looking for stocks."
User Presses Search Bar
Display Search History
Press an old stock
Search for stock by ticker number/name
Display Similar Stocks
Display Stock Info
Have sufficient funds?
No
Disable Buy Button
Own positions of stock?
No
Disable Sell Button
Yes
Enable Sell Button
Yes
Enable Buy Button
User Sells Stock
User Buys Stock
Display number of positions and funds gained
Display number and total cost
Confirm or Back?
Confirm
Is User sure?
Yes
Transaction Sent
No
Was User Buying or Selling?
Buying
Selling

Figure 3: Decision Tree for Market Orders

3.1.3 Functional Requirements

REQ-1: Users should be able to search for a publicly traded stock based on its ticker symbol or its company's name.
REQ-2: Users should be able to view a stock's current market value and when the market value was last updated.
REQ-3: Users should be able to buy a stock up to their current liquid funds in their investing account, and be notified if they do not have enough for the amount they intend to buy.
REQ-4: Users should be able to sell a position at market price during the time of confirmation.
REQ-5: Users should be able see the amount they will be charged for a market order.

3.2 User Login

3.2.1 Description/Priority

This is a high priority feature in order for users to safely access our application and access their sensitive information on the Bank of Jarvis' database. This feature is a part of our need to enforce user security in this application.

3.2.2 Stimulus/Response Sequences

1. If a user first opens the application, they will be directed to the login page.
2. If a user has a bank number and password, they can input them on their respective fields and press the "Login" button to log into their account.
3. If a user inputs incorrect credentials, they will be notified that either the username or password is incorrect, and to try again.
4. If a user inputs their credentials successfully, they will be notified that they will need to input their 2FA code, as well as different methods to receive a code.
5. If a user selects their preferred method, they should be able to input this code into a numeric field and log into their account.
6. If a user presses a "Log Out" button while they are logged in, they will be given a confirmation prompt asking if they wish to log out.
7. If a user presses "Yes", they will be logged out of their account and returned to the Bank of Jarvis bank page.

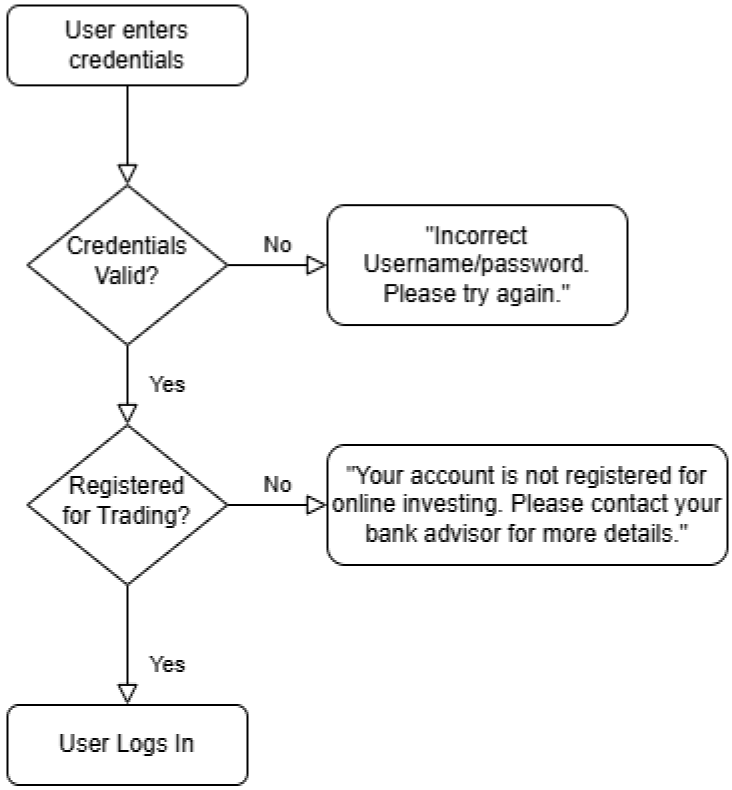

Figure 3: Decision Tree for Login Screen

3.2.3 Functional Requirements

REQ-1: Users should be able to input their banking number and password in order to log into their account.
REQ-2: Users should be able to receive and input a code from a 2FA notification using a trusted Bank of Jarvis 2FA authenticator.
REQ-3: Users should be able to log out of their accounts after logging in.
REQ-4: Users should be made aware that they will need to register for an account in person at their local Bank of Jarvis service branch.

3.3 User Portfolio

3.3.1 Description/Priority

This is a medium priority feature as it does not break the application's functionality, but will be important for the application's user experience and gives them the ability to view their current positions.

3.3.2 Stimulus/Response Sequences

1. If a user presses a “Profile” icon, they will be directed to a page with their profile information, current funds, and a condensed list of their stock portfolio.
2. If a user presses a “Portfolio” icon, they will be directed to a page with a detailed overview of all their stock positions in their account, and a breakdown of their portfolio’s valuation in a visual format.
    a. If a user presses on a stock, they will be directed to see detailed information of the stock, exactly like in 3.1 Stock Trading.
3. If a user presses a “Transactions” icon, they will be directed to a page with a history of their market orders and the number of stocks and dollar values spent or gained from every transaction.
    a. If a user presses a “Filter” icon, they will be given options to filter based on ticker number, date, dollar value, and number of stocks.
    b. If a user presses confirm, they will see an updated transactions list.

3.3.3 Functional Requirements

REQ-1: Users should be able to view their portfolio and all their current stock positions

REQ-2: Users should be able to view a history of their transactions.

REQ-3: Users should be able to see if a stock position is complete, or still being processed.

REQ-4: Users should be able view their personal information stored on the Bank of Jarvis’ database.

REQ-5: Users should be able to see the current valuation of their portfolio and how it is broken down.

REQ-6: Users should be notified that funds must be deposited into their account from their banking app, and be given steps on how to do so.

4. External Interface Requirements

4.1 User Interfaces

The interface should be geared towards simplicity, and accessibility in order to ensure new users know where to go in order to navigate to any given page. As a result, a navigation bar must be present throughout the main layout, with pop-ups existing if the user needs to be notified of any information, or confirm an important decision (i.e. placing a market order).

In addition to this, users should be given a walkthrough of the basic features of the app's landing page, to let them know which page to navigate to in order to perform a given action. A help icon should exist at every stage of the app to give users an idea of what they can do at a given page.

A user should be able to navigate to market orders and stock portfolios from anywhere in the application, and be able to easily log out or look for help.

If a transaction fails or the internet connection is lost, a pop-up message should display the failure that occurred.

Keep fonts to size 12 and above, and use easily readable font families, such as Calibri or Roboto, to minimize misreadings. Buttons should be big and clearly enabled or disabled, depending on context. Reasons for disabled buttons should be displayed as well. The navigation bar should be visible at any stage of the app.

4.2 Hardware Interfaces

In order for this application to be as widely accessible for the Bank of Jarvis customers, the application will be served as a web platform and a mobile application. The web platform will be served with HTML and accessible with any device that can run a web browser. Users are expected to use keyboard and mouse/trackpad, as well as mobile touch support.

The mobile applications will be served as .apk and .ipa files that users can download from their phone's app store. Both platforms will communicate with the Bank of Jarvis' remote database with an exposed endpoint and API calls.

The web application will be designed with desktop user interactions in mind, while the mobile applications will be designed for touch-based interactions. This requires the team to develop two sets of UI experiences in order to provide a comfortable experience.

4.3 Software Interfaces

This application is primarily a web and mobile interface for the Bank of Jarvis' database servers. They will be responsible for sending REST API requests to the database's endpoints, representing actions such as buying and selling stocks, and retrieving stock portfolio information.

Since web experiences differ from mobile experiences in their input devices, they require different user experience layouts geared towards their primary input methods. The web platform will serve as an HTML file and should be easily accessible on the Bank of Jarvis web domain. In order to focus on the application development, and not on distribution and maintaining the web server, the server will be set up on the cloud. The team should assume the web platform will be run with a web browser running a Windows, Mac, Android, or iOS operating system, thus providing both desktop and mobile support.

Additionally, the application should also be available as a mobile application, distributed as .apk and .ipa files for Android and Apple devices, respectively. These apps will require separate approval from the Google Play Store and the Apple App Store before they can be put on the app store for distribution.

4.4 Communications Interfaces

This application first and foremost will require a stable internet connection of at least 5 megabits per second and a device capable of running a web browser or mobile application. This will allow the application to serve as an interface for the Bank of Jarvis' database server. Communications will be limited to only allowing the user to log in, create market orders, and view limited information on their account. This is done by setting up endpoints with the Bank of Jarvis to ensure REST API requests can be processed. As such, communication between the server and the application will be done with HTTP request and response protocols. Any request that requires a user to be logged in will require a temporary session hash created by any successful login and have its JSON payload tokenized with an asymmetric key for security.

All packets that the application sends should be at most 50 kilobytes for CRUD operations, with a near-instantaneous response round-trip time. When retrieving a user's stock portfolio information, or when a user searches for a publicly traded stock, response times should be within 2-5 seconds, and be approximately 5 megabytes. Payloads from the database must also be sanitized of any Bank of Jarvis' private data.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

As the application should only be responsible for sending and receiving data from the Bank of Jarvis' database, the requirements will only focus on the user experience. In order to accommodate users who are not used to web and mobile applications, the application performance requirements should focus on clear visual feedback. This means the application should implicitly tell the user what it is doing and what the user should be doing on any given page.

For example, every button should be labelled and visually distinct from the rest of the page. When clicked, they should take no more than 2 seconds to generate a response. This response should tell the user what the application is doing: whether it is loading, waiting for a response from a database, or redirecting the user to a different page. This is to minimize confusion and to give new users a brief understanding of what the application is doing from their input.

For requests that do require the database, such as logging in or making a market order request, the application should respond within a minute and with a clear message of success or failure. That message should always be accurate and easily understood, as these transactions can potentially involve large sums of money.

5.2 Safety Requirements

Users are financially responsible for stock transactions made on the application on their own and are responsible for the risks therein, such as financial losses. Each transaction in the application will be accompanied by a confirmation in order to ensure users are given an opportunity to cancel their request before it is sent. In addition, all transactions and requests between the Bank of Jarvis database and application must be properly encrypted.

Users are also responsible for keeping their bank access number and passwords safe outside the application, and they are expected to be consistently reminded not to share their 2FA code with anyone. This should be done whenever the 2FA code is sent.

The Bank of Jarvis is responsible for ensuring their database follows ACID protocol, and transactions will not get lost in transit, as this poses a financial loss to the user and a severe loss of trust for the bank. Users should receive a confirmation as soon as the database successfully receives the purchase request. Both the application and the database will be responsible for ensuring that suspicious and malformed transactions are rejected, such as transactions that cost more than a user's liquid assets, or selling more than the total positions.

5.3 Security Requirements

The user will need to input their bank access number and a personal password in order to access the application's main services. Since most of the data is sensitive, all data communications with the application must be encrypted with an asymmetric key generated by the database or a hash for the bank account number and password. The asymmetric key must be generated per session to ensure safety and have a self-imposed expiration. As the application will be using more modern operating systems, there will be a lower security risk, but the users will be expected to keep up to date with their own operating systems.

In order to protect users from hackers, 2FA is required in order to access the application as well. The Bank of Jarvis will be responsible for providing its supported 2FA services for integration into the application, and will be required to be set up in-person when a customer creates their account for online investment. This sensitive data will not be accessed by the user through this app to maintain a need-to-know policy.

5.4 Software Quality Attributes

The primary objective of this application is to ensure users will be able to quickly place market orders and expect a clear confirmation once the request has been received. As such, the software must be reliable and usable first and foremost. To measure those qualities, if the software is able to return a successful or unsuccessful response within a minute of sending a market order 100% of the time, we can declare that the software is highly reliable. If at least 80% users are able to search up a ticker number for a specific stock and make a market order in less than three minutes, we can declare the software to be usable.

6. Other Requirements

Appendix A: Glossary

2FA — Two-factor authentication; a common method to secure personal accounts by adding an additional constantly changing code that is received by only the intended user.

REST API — “Representational State Transfer Application Programming Interface”. A web development standard for communication between a client and server over the internet. If adhered, it allows for safe and effective communication between the two without exposing more information than is necessary for both sides to function.

IoT — “Internet of Things”. An expression that represents applications or systems that require a network in order to function.

Malicious Actors — Black-hat hackers, or people who act with the express intent to harm or take advantage of a system for personal or monetary gain.

Transactions — A logical unit of work for a database, typically representing a read, write, update, or delete action that a database is expected to perform.

Batch store — A method for databases to perform small individual data stores in parallel as large aggregated chunks.

Platform — A piece of hardware that can run an application.

Asymmetric Key — A one-way encryption method that allows a sender to safely send information, and ensuring only the receiver can read it. The receiver creates an encryption and decryption key and sends an encryption key over to the sender for use. The receiver then uses the decryption key in order to properly read the contents of a sent message.

Human-readable format — Information that is presented such that a human will be able to read and understand the context without external tools or help.

Appendix B: Issues List

- Who will be responsible for creating and exposing datapoints to access the Bank of Jarvis database?
- What will be the framework to be used during the development of the web and mobile application?

Appendix C: References

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